Did voting within neighborhoods become more homogeneous?

Measuring political variation within Turkey's neighborhoods using about 919,000 ballot-box records from five parliamentary elections, 2011–2023.

1. The question

In 2020, Taşkın and Özer examined election results in a gated residential complex in Tuzla ("Ayıran Duvarlar, Dönüşen Sandıklar," Liberal Düşünce Dergisi). The smallest unit in the published results is the sandık, or ballot-box unit, but the data do not identify which side of a wall its assigned voters live on. The researchers interviewed local party organizations to determine which ballot-box numbers belonged to the complex. Their study showed that vote distributions could differ sharply between boxes separated by a single wall.

This paper applies the same type of measurement nationwide over twelve years: did ballot boxes within the same neighborhood become more similar over time, or did the differences between them grow? The question is related to the American "Big Sort" literature. The contribution here is to measure within-neighborhood ballot-box variation nationwide and longitudinally on a geographically comparable subset.

The short version:

Within neighborhoods that remained comparable across all five elections from 2011 to 2023, the vote shares of the three long-established parties (AKP, CHP, and MHP) became less dispersed across ballot boxes. Within this comparable subset, political variation declined modestly, with the clearest decrease for CHP.

Before reaching that conclusion, the analysis tested three potential sources of artifact. An initially opposite result for HDP/YSP did not survive support-floor checks and was rejected. These controls define how the results can be interpreted.

2. The data

The data are ballot-box-level results from five parliamentary (milletvekili) general elections: June 2011, June 2015, November 2015, June 2018, and May 2023. After cleaning, each election contributes roughly 174,000 to 199,000 usable records, about 919,000 in all. The registered electorate grows from 46 million in 2011 to 56 million in 2023.

Each record contains province (il), district (ilçe), neighborhood (mahalle), and ballot-box number. A mahalle is an official administrative neighborhood and is the key unit in this analysis. It typically contains a few to a few dozen boxes, each serving several hundred voters assigned by address. Individual votes are never observed, but the records show how evenly or unevenly party support is distributed across boxes in the same neighborhood. Studies using only neighborhood totals cannot calculate this internal variation.

Two limits of the data, stated up front:

- For 2011, 2018, and 2023, the neighborhood is the finest geographic key. Only the two 2015 elections carry school/building names. Boxes within a neighborhood can be compared to each other, but not placed on a map more finely than their neighborhood.
- All results are areal measures describing ballot boxes and neighborhoods, not individuals. They do not establish that any voter changed their preference, party, or address.

Two cleaning choices are particularly important. First, standard lowercasing can break matches involving the dotted and dotless Turkish İ/ı. All place-name joins therefore use a tested Turkish-aware normalizer. Second, vote shares are calculated against reported valid votes (geçerli oy), not the sum of major-party columns; a residual column retains votes for smaller parties and independents. Using only major-party totals as the denominator would systematically inflate their shares. Alliance votes in the 2018 and 2023 files remain in separate columns and are not added to party totals.

3. The measure

For each neighborhood, election, and party, the calculation measures how much the party's vote share varies across ballot boxes in that neighborhood.

Each box's party share (party votes ÷ valid votes) is calculated, followed by the box-size-weighted standard deviation across boxes in the neighborhood. If AKP receives 40% at every box, dispersion is zero regardless of the neighborhood's overall political leaning. If it receives 20% at half the boxes and 60% at the other half, within-neighborhood variation is high.

Raw dispersion scales with a party's support and is therefore insufficient on its own for trend analysis. If support falls from 45% to 10%, absolute dispersion can shrink mechanically even if the geographic distribution is unchanged. The principal statistic is therefore the coefficient of variation (CV): dispersion divided by the party's mean share in the neighborhood. CV expresses variation relative to the size of local support. National summaries weight neighborhoods by registered voters, so a 300-voter village and a 40,000-voter urban neighborhood are not treated as equal observations.

Calculation and aggregation

For box `b` in mahalle `m`, let `p_b` be the party's votes divided by valid votes and let `w_b` be the larger of its registered-voter and valid-vote counts. The mahalle mean and absolute dispersion are:

`mean_m = sum_b(w_b p_b) / sum_b(w_b)`

`SD_m = sqrt(sum_b(w_b (p_b - mean_m)^2) / sum_b(w_b))`

The CV is then `CV_m = SD_m / mean_m`. SD measures the absolute-scale dispersion of ballot-box vote shares around the mahalle mean and has the same units as vote share. CV is a dimensionless ratio that divides this dispersion by the party's mean local support. CV is the principal trend measure; SD is also reported as a directly interpretable measure of absolute dispersion.

For the national line charts, the reported value is the registered-voter-weighted mean of the individual mahalle CVs: `sum_m(V_m CV_m) / sum_m(V_m)`, where `V_m` is the mahalle's registered electorate. It is therefore not a single countrywide CV recomputed after pooling all boxes. The headline lines use the full-chain stable-neighborhood subset (condition 4) described below.

The district maps use a different descriptive aggregation: the registered-voter-weighted mean neighborhood SD divided by the registered-voter-weighted mean neighborhood party share. The interactive map can also display the weighted mean SD alone. Districts below 10% mean party support are masked because a small denominator makes CV unstable. The map is descriptive and does not provide independent evidence for the national trend.

4. Three potential artifacts and their controls

The initial calculation takes the registered-voter-weighted election mean of neighborhood CVs and shows a decline over time. Three potential artifacts were tested to determine whether measurement or sample composition produced that trend.

Control 1: the changing party system. Fragmentation of the party system can lower major-party vote shares, mechanically reducing raw SD. The decline remains when CV is used instead: voter-weighted national CV falls from 2011 to 2023 for AKP, CHP, and MHP, the parties present throughout (0.165→0.134, 0.325→0.177, and 0.351→0.284 respectively in the uncontrolled series).

Control 2: ballot-box counts. The number of boxes per neighborhood changed over time, and measured dispersion was related to box count. The relationship was material in 11 party-election cells, so box-count stability became an explicit control condition.

Control 3: boundary and name changes. Neighborhood identifiers do not carry over continuously between elections. Only 42% of 2011 neighborhood names reappear in the June 2015 data. The 2012–2014 metropolitan-municipality reform coincides with the reclassification of many villages as neighborhoods; the data show the change but do not independently establish its cause. After 2015, name-match rates are about 85–87%, but in some periods the matched neighborhoods contain only about half of registered voters. Directly comparing all available neighborhoods at each election can therefore change sample composition substantially.

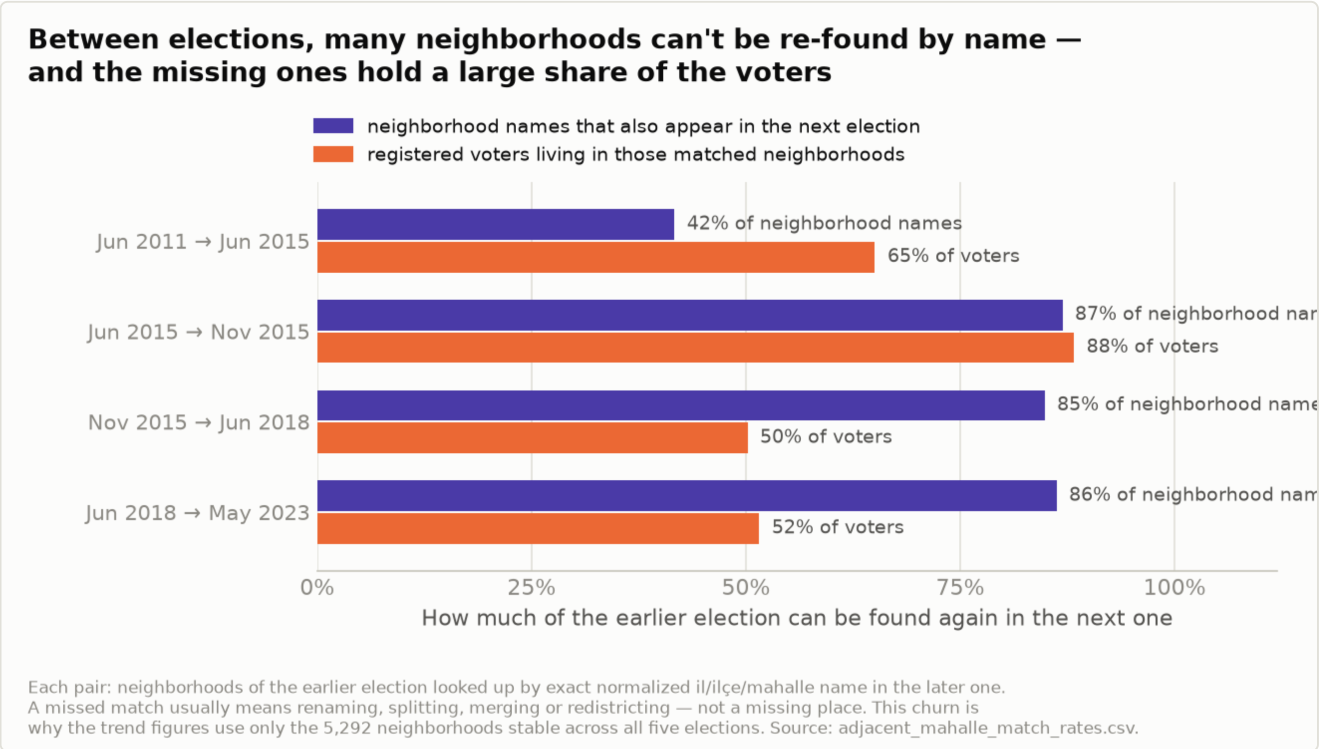

Figure 5. Adjacent-election continuity: share of neighborhood names re-found in the next election (violet) and of registered voters living in the matched neighborhoods (orange).

Four control specifications address these problems. Geographic continuity and ballot-box counts are first tested separately; the strict final specification combines them. This **full-chain stable-neighborhood subset (condition 4)** includes neighborhoods whose province/district/`mahalle` key exists in all five elections and whose box count remains stable: at least 2 boxes in every election, with a maximum/minimum ratio ≤ 1.5 or an absolute range of at most 2. It leaves **5,292 full-chain stable neighborhoods**, covering about 9.5 million registered voters in 2011 and 11.3 million in 2023. Coverage is deliberately narrowed in exchange for comparability. A province-aggregated series provides an independent directional check.

5. The finding

On the strict full-chain stable-neighborhood subset (condition 4), voter-weighted within-neighborhood CV:

PARTY	2011	2023	DIRECTION
AKP	0.158	0.127	decline
CHP	0.310	0.165	strong decline
MHP	0.339	0.275	decline

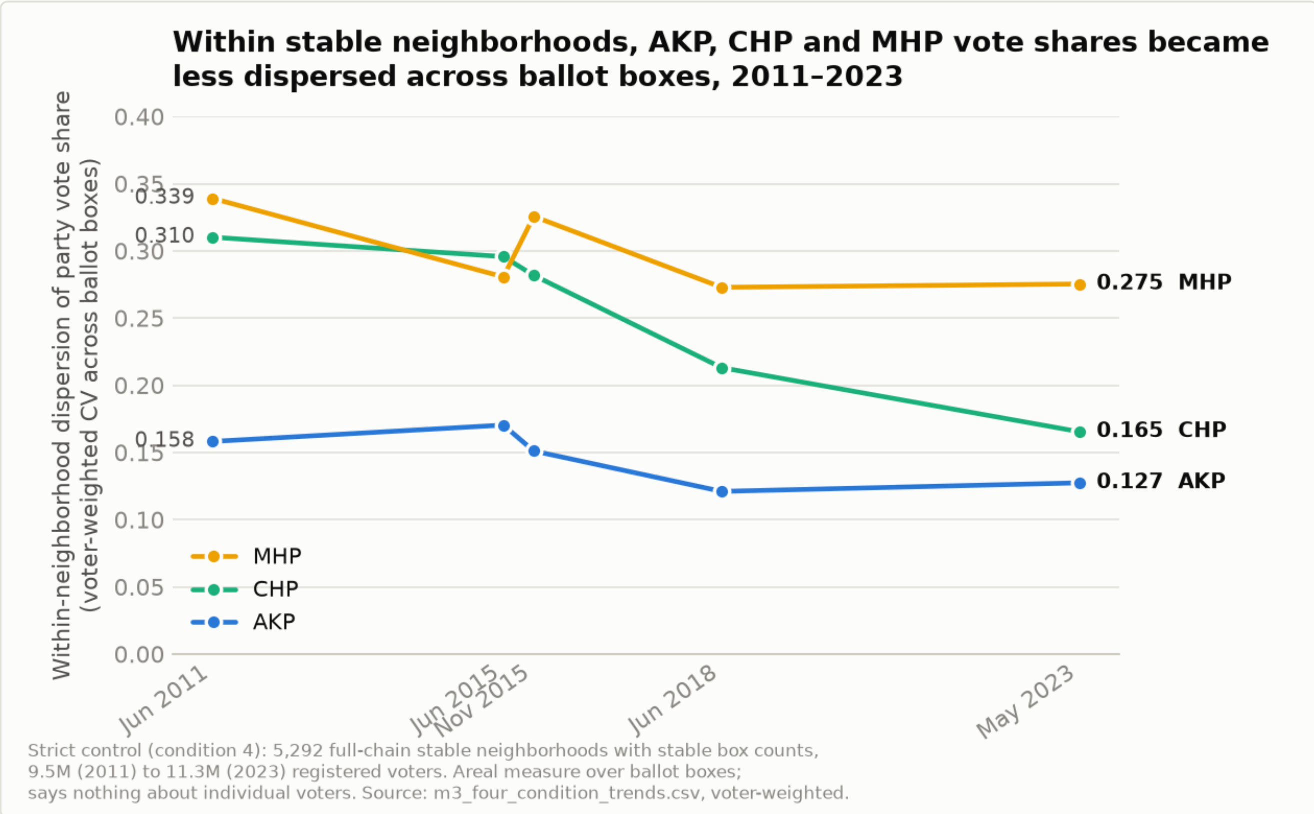

Figure 1. Within-neighborhood dispersion (voter-weighted CV) for AKP, CHP, MHP on the 5,292 full-chain stable neighborhoods of condition 4.

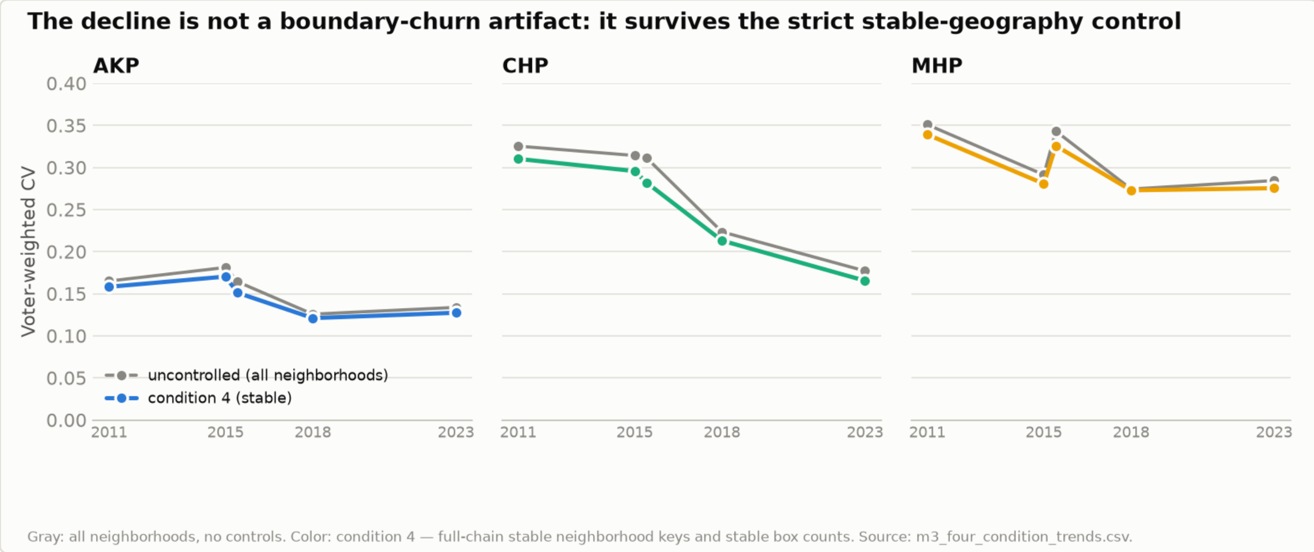

Figure 3. Uncontrolled national series (gray) versus condition-4 stable-geography series (color). The decline survives the strictest control.

Three things about this result.

The controls change the estimates but not their direction. The uncontrolled and condition-4 series are not identical, yet both decline for all three parties. This indicates that boundary/name changes and shifting box counts do not by themselves account for the observed decline.

The metric does not move mechanically in one direction. MHP's raw within-neighborhood dispersion rises from 2011 to June 2015 (0.033→0.034) before falling in later elections. This non-monotonic path is a useful internal check that the measure does not reflect only a universal mechanical compression.

The magnitude is modest. CHP's within-neighborhood CV roughly halved over twelve years, while AKP's and MHP's declined by about a fifth. The result shows reduced within-neighborhood political variation among established parties in the comparable subset. It does not support the broader claim that "Turkey homogenized."

6. The rejected HDP/YSP fragmentation result

The initial HDP/YSP series moved in the opposite direction. The party crosswalk uses HDP for 2015–2018 and Yeşil Sol/YSP for 2023. On the same stable subset, within-neighborhood CV rose from 0.524 in June 2015 to 0.630 in 2023, suggesting increased internal fragmentation unlike the pattern for AKP, CHP, and MHP.

That interpretation fails the support-floor check. A coefficient of variation can become very large as mean party support approaches zero: at a 1% mean share, small absolute differences between boxes produce large relative dispersion. If the apparent rise were a robust neighborhood-level fragmentation trend, it should remain after restricting the sample to neighborhoods where the party reaches a defined support level. Instead, the direction reverses:

NEIGHBORHOODS INCLUDED (CONDITION-4 SUBSET)	JUN 2015 CV	2023 CV	DIRECTION
All	0.524	0.630	rises
Party ≥ 5% support	0.341	0.243	declines
Party ≥ 10% support	0.268	0.168	declines
Party ≥ 20% support	0.157	0.101	declines

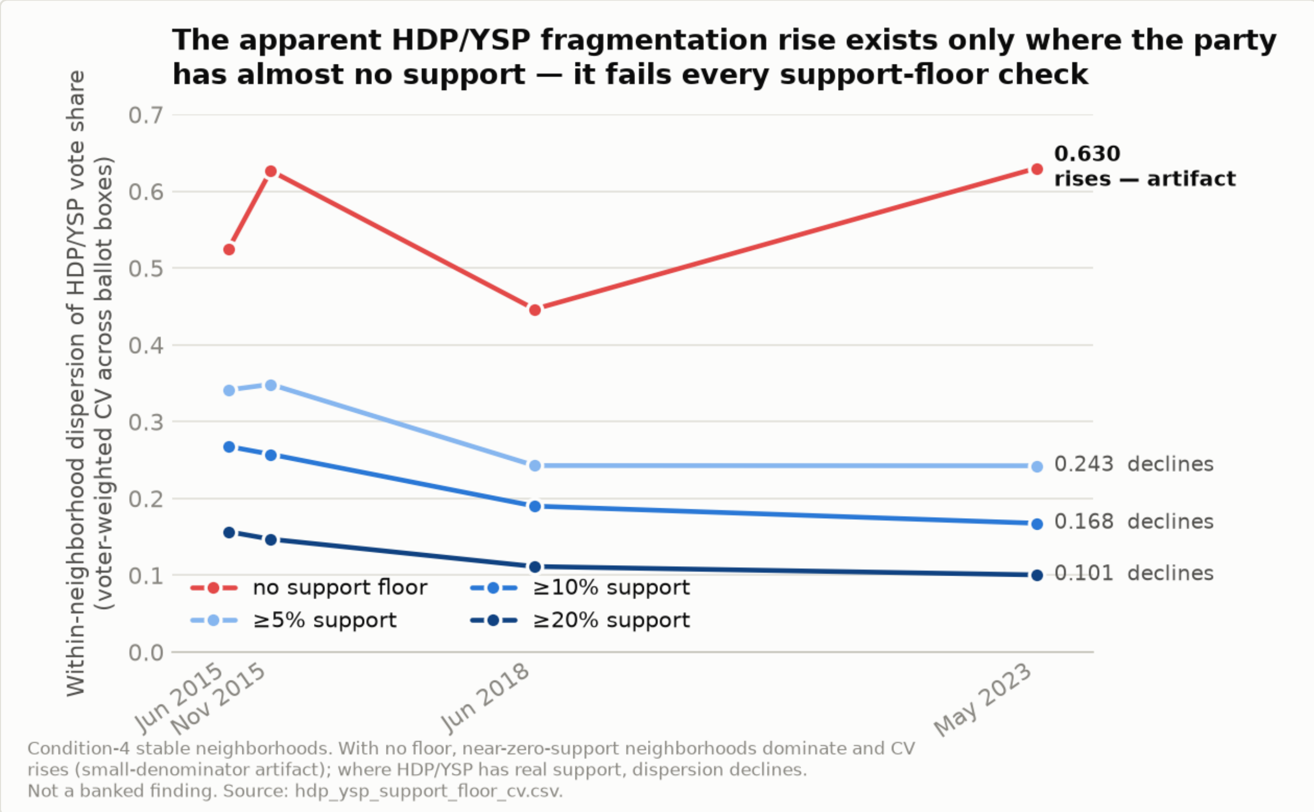

Figure 2. HDP/YSP within-neighborhood CV on the condition-4 subset, with and without support floors. The rise exists only with no floor — a small-denominator artifact, not a finding.

The rise disappears and reverses after support floors are applied. This is not treated as an independent finding of HDP/YSP homogenization. Rather, it shows that the no-floor national increase is driven largely by the small-denominator effect in near-zero-support neighborhoods. The province-aggregated control series also shows no increase, while turnout and composition checks in the southeast provide no alternative substantive explanation.

The paper therefore makes no claim of nationwide growth in within-neighborhood HDP/YSP fragmentation. The case demonstrates why support floors are necessary when interpreting CV for regionally concentrated parties.

7. Examples of high within-neighborhood variation

In addition to the national trend, party-neighborhood-election cells can be ranked by internal dispersion to identify extreme examples. This approach scans nationwide public data for the type of within-neighborhood variation that Taşkın and Özer examined through fieldwork at one residential complex.

To reduce noise from small boxes, the main list requires at least 4 boxes and at least 150 registered voters per box. The filtered top 100 overlaps the unfiltered top 100 by 81%, so the extremes do not consist only of small boxes. In Uzunziyaret Mahallesi (Siverek, Şanlıurfa) in June 2015, AKP received 12% at one box and 99% at another; the neighborhood had four boxes and 885 registered voters. In Kazımiye Köyü (Yalova), HDP ranged from 2% to 88% across five boxes, and the same place appears in the extreme list in November 2015 and 2018. Because boxes are assigned by address, these differences may be related to local social or spatial boundaries, but confirming that relationship requires local data or fieldwork.

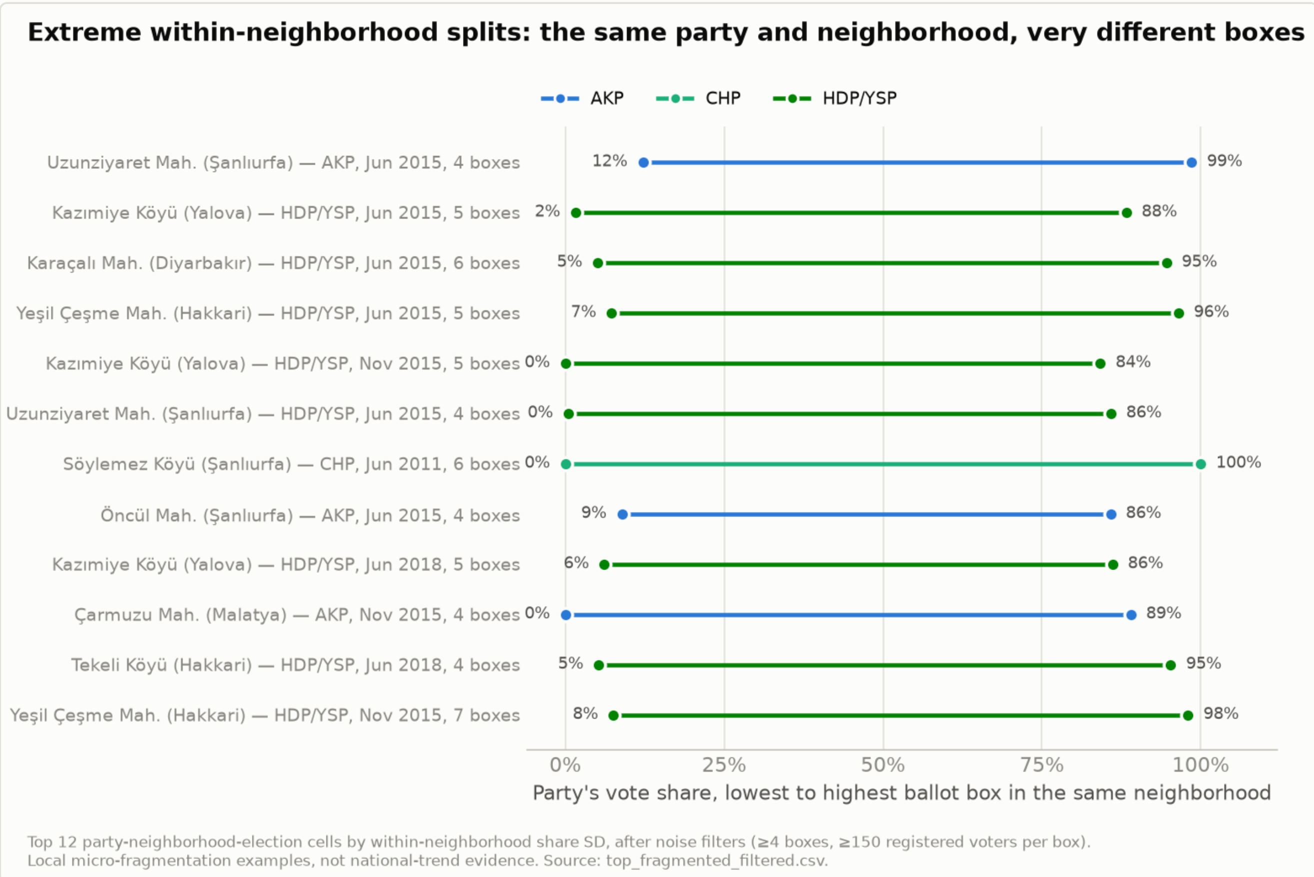

Figure 4. Party-neighborhood-election cells with the highest within-neighborhood variation after noise filters (≥4 boxes, ≥150 registered voters per box): lowest and highest ballot-box share in one neighborhood.

These are local examples, not trend evidence; a list of extremes is unrepresentative by construction. It illustrates what within-neighborhood dispersion means and identifies candidate areas for local investigation.

8. The map and its spatial scope

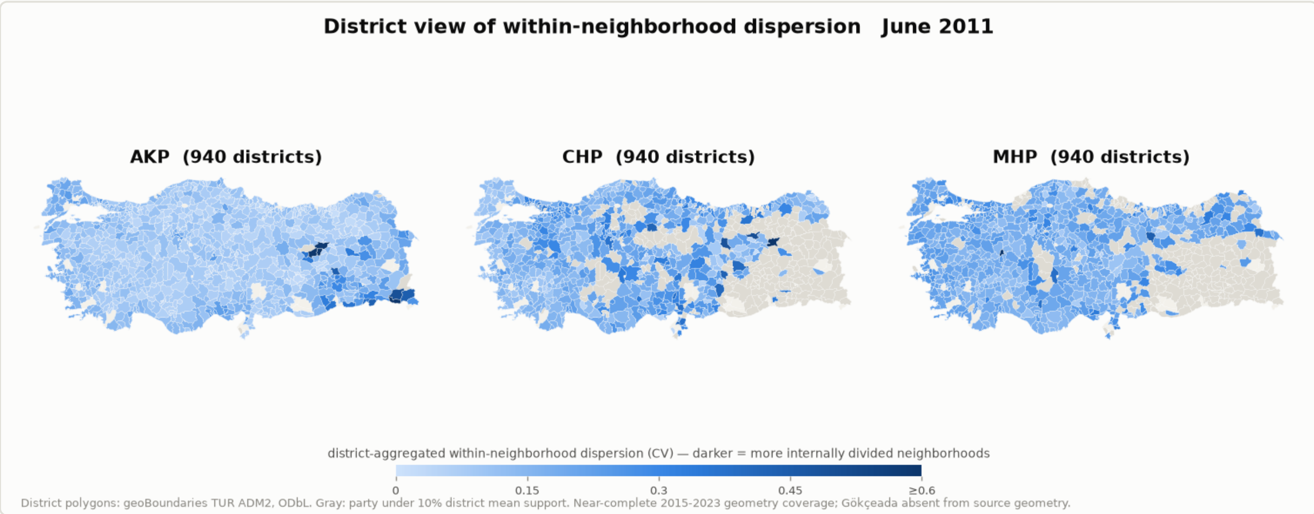

Figure 6a. District aggregation of within-neighborhood dispersion per party, June 2011 (first frame of the animation). Gray = under 10% district mean support or no party list.

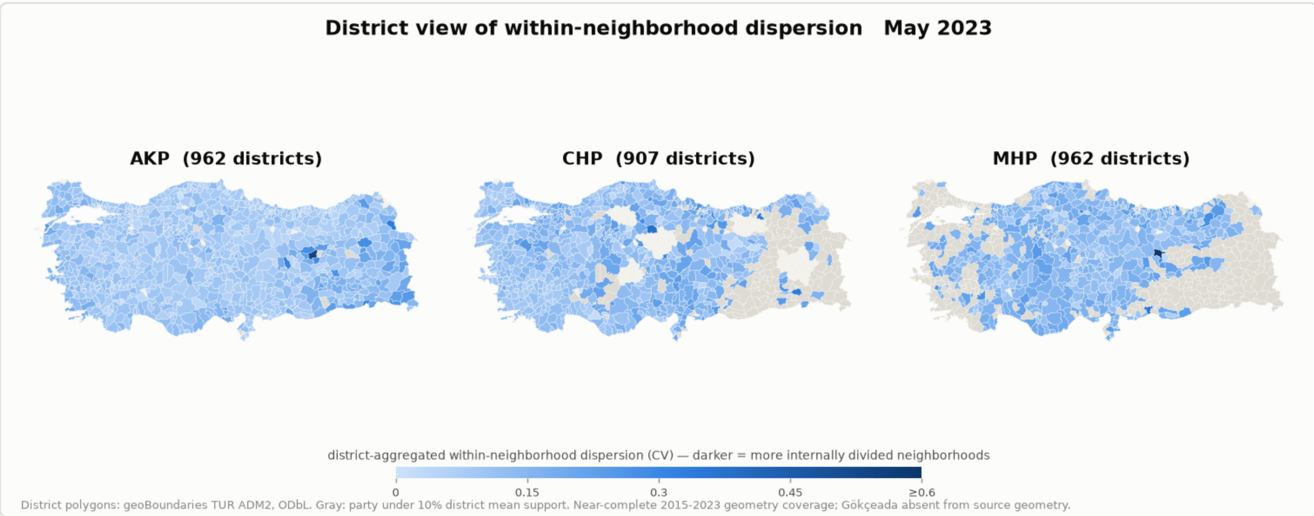

Figure 6b. The same district maps, May 2023 (last frame). The animated version steps through all five elections.

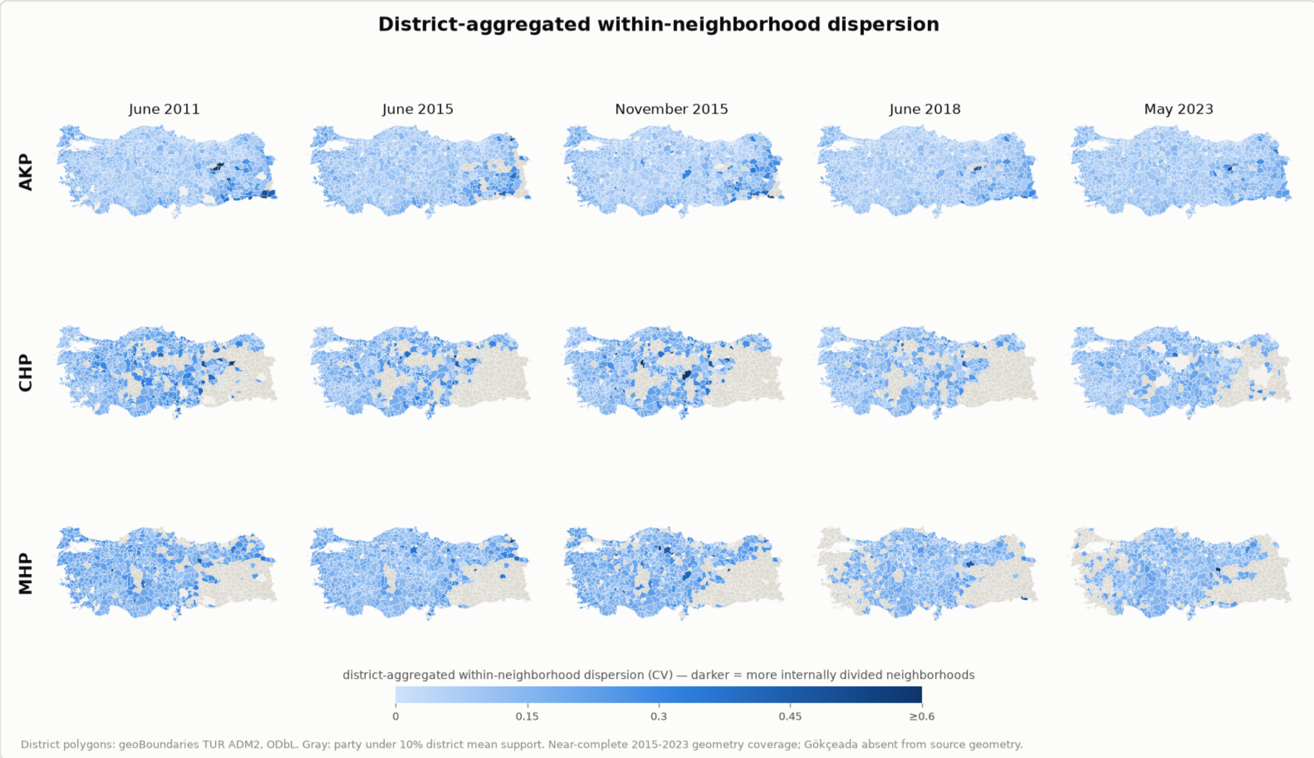

Figure 6c. All five elections as a static district grid — rows fade toward lighter (more internally uniform neighborhoods within districts).

A map shading every neighborhood by its internal variation would require national neighborhood boundary polygons, which are not available in this project (see §9). The most granular defensible filled map is therefore district-level: each ilçe polygon is shaded by the voter-weighted within-neighborhood dispersion of its neighborhoods, by party and election. District geometry comes from geoBoundaries' Turkey ADM2 polygons. The join is effectively complete for 2015–2023, with only Gökçeada missing from the source geometry; the 2011 map carries a stronger historical-boundary caveat.

At this resolution, the CHP panels lighten distinctly from 2011 to 2023. AKP remains mostly low, with some districts showing higher within-neighborhood dispersion, while the change for MHP is more limited. Districts are masked gray where mean party support is below 10%, because CV can appear high when its denominator is small. Districts without a usable party list or geometry are also gray. The support floor is a limitation on interpretation, not merely a web-map interface choice.

The map is a descriptive district aggregation of neighborhood-level dispersion. It visualizes a pattern consistent with the main finding but does not provide independent evidence for it.

The analysis also tests whether stable neighborhoods formed stronger geographic clusters with nearby neighborhoods as their internal boxes became more similar. Using OSM place-node centroids for the condition-4 subset, longitudinal FDR-screened LISA transition tests show no robust change in spatial clustering from 2011 to 2023 for AKP, CHP, or MHP. AKP shows endpoint transition structure under one tested weight rule (k=8), but not under the primary, k=6, or 10 km-cap rules; CHP and MHP show no robust start-end transition signal. The centroid join covers 4,253 of 5,292 stable neighborhood keys, 85.3% of registered voters in the 2023 stable subset, and about 90.7% of southeast voters in that subset. This coverage reduces concern that missing centroids primarily drive the result, but it does not remove the concern because unmatched neighborhoods are not random.

Gençkaya and Bakırcı (2023) found a distinct electoral geography when clustering the 2018 election at district level. The longitudinal LISA test here shows that the decline in within-neighborhood variation was not accompanied by a detectable clustering change under the tested resolution and specifications. This is an exploratory negative result, not proof that spatial structure did not change. HDP/YSP is considered separately in this layer because its fine-resolution results are more sensitive to support floors and spatial-weight choices.

9. Scope and limitations

These limitations are part of the interpretation.

- **Areal, not individual.** The measure describes variation among ballot boxes within neighborhoods. It cannot distinguish residential movement, registration changes, local political change, generational turnover, or other mechanisms. It supports no individual-level claim.
- **Modest magnitude.** Within-neighborhood CV falls for three established parties and most strongly for CHP. The result does not imply a broad national transformation.

- **The stable subset is not a random sample.** Condition 4 contains 5,292 neighborhoods whose names and box counts remain stable. Continuity checks show that excluded neighborhoods are more populous and metropolitan on average: in some periods, matching neighborhood names contain only about half of registered voters. Coverage is narrowed for comparability. The uncontrolled series, which agrees in direction and includes about 19,000–24,000 neighborhoods per election, is shown alongside it.
- **Stable controls match neighborhoods, not ballot boxes.** Within a stable neighborhood, box numbers and the voter groups they cover can change between elections; the panel is neighborhood-level, not box-level.
- **Resolution limits are unresolved (MAUP).** All dispersion is measured within administrative neighborhood units. The 2015 elections' school names would allow a finer-resolution recomputation to test how much the unit choice matters; that calibration has not been done.
- **This study extends prior research.** Turkish voting has been studied spatially at province, district, and single-province neighborhood levels (Yiğit and Sezgin, 2019; Gençkaya and Bakırcı, 2023; Ölmez and Elmastaş, 2024). Taşkın and Özer (2020) used fieldwork to examine ballot-box-level variation in one gated residential complex. This study extends that literature by measuring within-neighborhood ballot-box dispersion nationwide and longitudinally on a matched stable-geography panel with explicit artifact controls.

10. Where this could go next

Three natural extensions are available: geocoding the 2015 school names to measure the effect of neighborhood aggregation; adding local context to cases with high within-neighborhood variation; and using registration and turnout dynamics to examine whether the observed change is associated with population movement or with change among more stable populations. These extensions are not required for the present result, but the panel built here provides a foundation for them.

References

- Gençkaya, Ö. F., and Bakırcı, E. (2023). "Türkiye'de Seçim Coğrafyası: 24 Haziran 2018 Seçimleri Üzerine İlçe Temelli Bir Analiz." Yasama Dergisi, 46, 73-112.
- Ölmez, İ., and Elmastaş, N. (2024). "Seçimlerin Mekânsal Analizi: Hatay İli Örneği." Turkish Studies, 19(2), 787-806. https://doi.org/10.7827/TurkishStudies.75358
- Taşkın, B., and Özer, B. (2020). "Ayıran Duvarlar, Dönüşen Sandıklar: Kapalı Konut Siteler ve Seçim Sonuçlarına Etkileri." Liberal Düşünce Dergisi, 25(100), 137-167. https://doi.org/10.36484/liberal.818092
- Yiğit, H. H., and Sezgin, A. (2019). "Türkiye'de Milletvekili Genel Seçimleri Üzerine Bir Mekânsal Bağımlılık Analizi." Süleyman Demirel Üniversitesi Sosyal Bilimler Enstitüsü Dergisi, 35, 1-28.

Appendix: provenance

The following files support the reported values and generate the figures:

- Cleaned data: `outputs/cleaned_mv/` (five elections; conformance reports in the same directory). Cleaning contract and tests: `CLEANING_SPEC.md`, `tests/test_cleaning_contract.py`.
- Condition-4 trend and coverage (§5, Figs 1 & 3): `outputs/m3_boundary_boxcount_control/m3_four_condition_trends.csv`, `m3_condition_coverage.csv`; control definitions in `m3_audit.json`.
- Uncontrolled CV check and box-count correlations (§4): `outputs/pre_m3_artifact_checks/artifact_check_verdict.json`.
- HDP/YSP resolution (§6, Fig 2): `outputs/hdp_ysp_resolution/hdp_ysp_support_floor_cv.csv`, `hdp_ysp_verdict.json`.
- Fragmented-neighborhood list (§7, Fig 4): `outputs/pre_m3_artifact_checks/top_fragmented_filtered.csv`.
- Continuity diagnostics (§4, Fig 5): `outputs/cleaning_audits/adjacent_mahalle_match_rates.csv`.

- Map (§8, Fig 6): `outputs/geo_join/district_map_values.csv`, geometry from local `geoBoundaries-TUR-ADM2-all.zip`, join audit `outputs/geo_join/geoboundaries_adm2_join_coverage.csv`, rendered by `scripts/make_district_choropleth_map.py`.
- M2 longitudinal LISA null (§8): `docs/m2_lisa_transitions_status.md`, `outputs/m2_lisa_transitions/`, `outputs/m2_closeout_checks/`, `scripts/m2_lisa_transitions.py`, and `scripts/m2_closeout_checks.py`.
- Figures: `scripts/make_findings_figures.py`, `scripts/make_district_choropleth_map.py`, and `scripts/make_homogenization_map.py` → `outputs/figures/`.

The scope of the claims is consistent with the entries in `FINDINGS_BANK.md`.